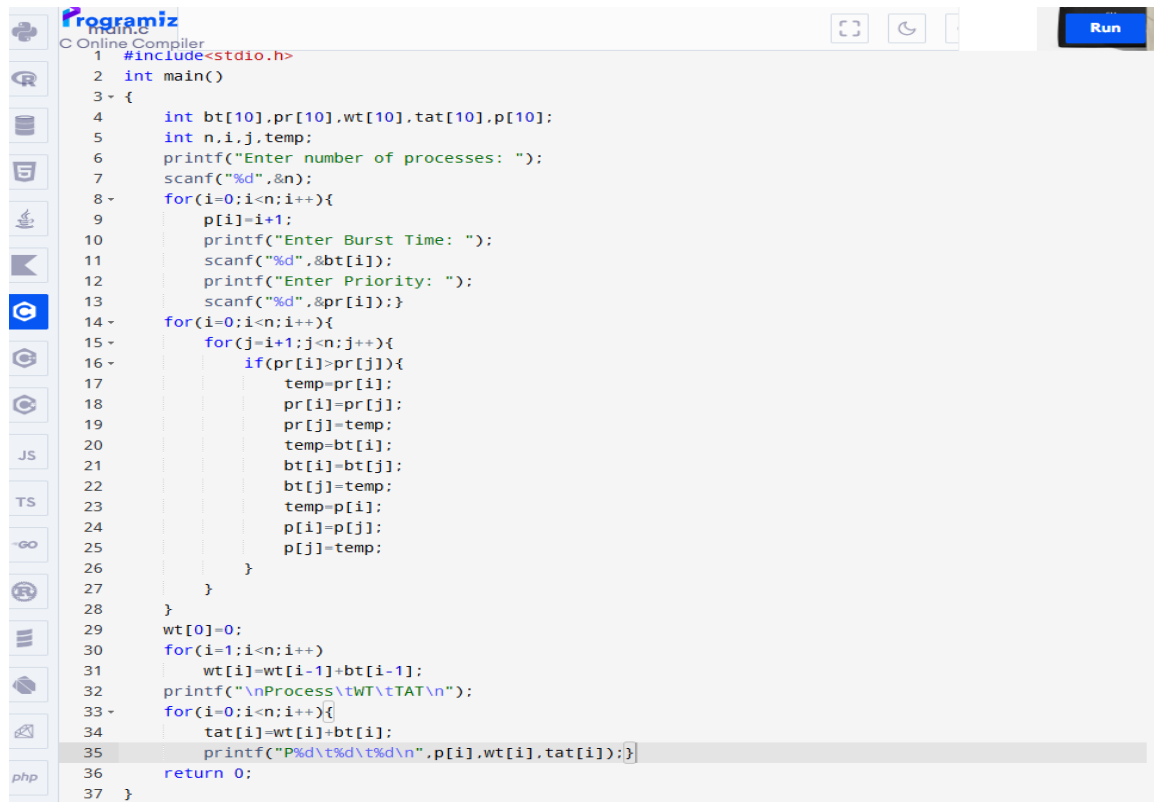


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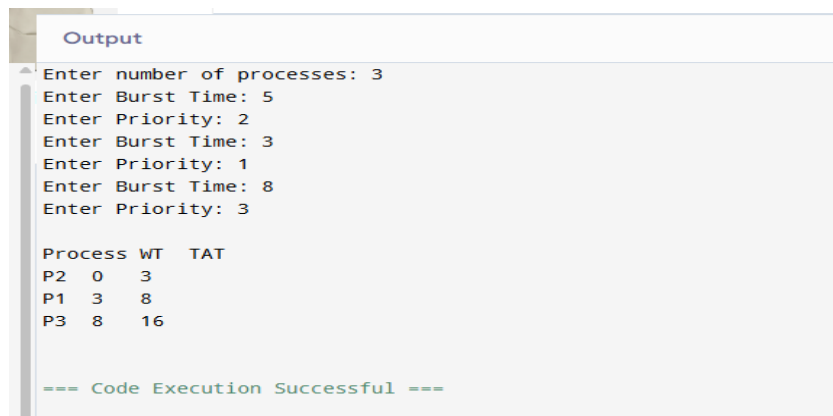
5. Construct a scheduling program with C that selects the waiting process with the highest priority to execute next.

Code:



```
1 #include<stdio.h>
2 int main()
3 {
4     int bt[10],pr[10],wt[10],tat[10],p[10];
5     int n,i,j,temp;
6     printf("Enter number of processes: ");
7     scanf("%d",&n);
8     for(i=0;i<n;i++){
9         p[i]=i+1;
10        printf("Enter Burst Time: ");
11        scanf("%d",&bt[i]);
12        printf("Enter Priority: ");
13        scanf("%d",&pr[i]);
14        for(i=0;i<n;i++){
15            for(j=i+1;j<n;j++){
16                if(pr[i]>pr[j]){
17                    temp=pr[i];
18                    pr[i]=pr[j];
19                    pr[j]=temp;
20                    temp=bt[i];
21                    bt[i]=bt[j];
22                    bt[j]=temp;
23                    temp=p[i];
24                    p[i]=p[j];
25                    p[j]=temp;
26                }
27            }
28        }
29        wt[0]=0;
30        for(i=1;i<n;i++){
31            wt[i]=wt[i-1]+bt[i-1];
32        }
33        printf("\nProcess\tWT\tTAT\n");
34        for(i=0;i<n;i++){
35            tat[i]=wt[i]+bt[i];
36            printf("P%d\t%d\t%d\n",p[i],wt[i],tat[i]);
37        }
38        return 0;
39    }
```

Output:



```
Enter number of processes: 3
Enter Burst Time: 5
Enter Priority: 2
Enter Burst Time: 3
Enter Priority: 1
Enter Burst Time: 8
Enter Priority: 3

Process WT  TAT
P2  0   3
P1  3   8
P3  8  16

=== Code Execution Successful ===
```